

Case Study: ZeptoFresh – 15-Minute Food & Essentials Delivery

PG Diploma · AI-ML & Agentic AI Engineering · IIT Gandhinagar

Duration: 60–75 minutes

Submission: Submit on LMS with GitHub Repository Link

Company Background

ZeptoFresh operates **350+ micro-fulfillment hubs** across **25 Indian cities**, promising **15-minute delivery** of groceries and ready-to-eat meals.

The analytics team is performing **Exploratory Data Analysis (EDA)** on **110,000 order records** from **Q2 2025** before building a **late delivery risk prediction model**.

(a) Data Quality Diagnosis

Here are four distinct problems, each mapped to a specific observation:

Problem 1 — `delivery_time_mins = 0` (214 rows)

- **Type:** Data Entry Error / Impossible Value
- **Treatment:** These are physically impossible — a delivery cannot take 0 minutes. Flag and remove these rows, or impute using the median delivery time for the same `hub_id` and `order_hour` combination.

Problem 2 — `order_value_Rs = ₹2,95,000` (single bakery order)

- **Type:** Outlier (likely erroneous)
- **Treatment:** Cap using the IQR method or Winsorize at the 99th percentile. Cross-check against the actual order record — a genuine bulk B2B order should be separated from retail data before modeling.

Problem 3 — `prep_time_mins` has negative values

- **Type:** Structural / Data Entry Error
- **Treatment:** Negative prep time is logically invalid. Replace negative values with NaN, then impute using the median `prep_time_mins` grouped by `order_category` (since Bakery vs Grocery prep times differ significantly).

Problem 4 — `customer_rating` has 9,800 nulls and values of 0

- **Type:** Missing Values + Out-of-Range Values

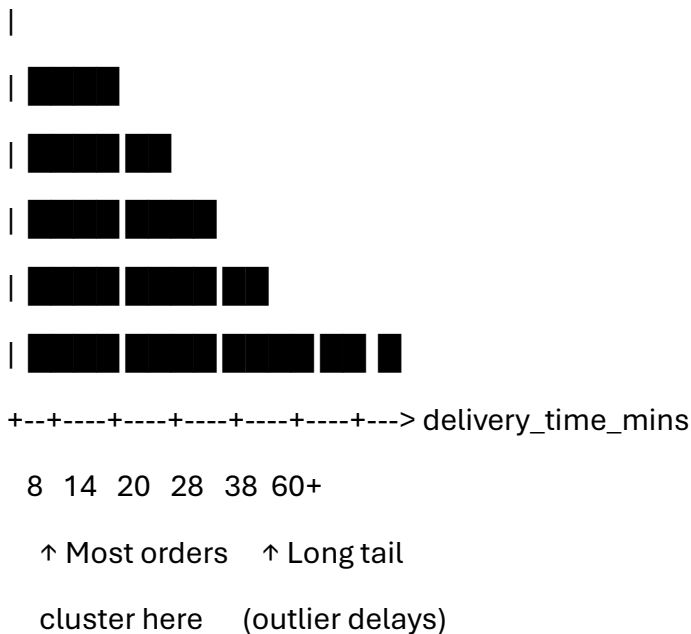
- **Treatment:** Ratings of 0 are outside the valid 1–5 scale — treat them as missing. For the nulls, do **not** impute with mean (it distorts sentiment analysis). Instead, create a binary flag `rating_given = 0/1` and leave nulls as-is, since a missing rating itself carries signal (customers who didn't rate may have had neutral experiences).

(b) Distribution Analysis

What does Mean > Median indicate? When mean (18.4) > median (14.2), the distribution is **right-skewed (positively skewed)**. A long tail of unusually high delivery times (up to 142 mins) is pulling the mean upward, while most deliveries cluster around 12–15 minutes.

ASCII Histogram:

Frequency



Which transformation and why? Apply a **Log Transformation** → `log(delivery_time_mins)`.

This compresses the long right tail, making the distribution approximately normal. Most ML models (Linear Regression, Logistic Regression) assume near-normality in features. Without this, extreme values like 142 mins disproportionately influence model coefficients.

⚠ Important: Since you have 0 values in this column, use `log(x + 1)` to avoid `log(0) = -∞`.

(c) Correlation Interpretation

The Product Manager's claim: "Late deliveries cause refunds."

What's logically incorrect? Correlation does not imply causation. An $r = +0.74$ tells you that late deliveries and refunds *move together*, not that one causes the other. The causal chain may be reversed, indirect, or driven by a third variable entirely.

What does $r = +0.74$ actually tell you? There is a **strong positive linear association** between `delivery_time_mins` and `refund_issued`. When one is high, the other tends to be high too — but the relationship could be coincidental, bidirectional, or mediated.

Two possible confounders:

Confounder	Explanation
<code>rain_flag</code>	Rain causes both delayed deliveries AND damaged packaging → leading to refunds independently
<code>order_category</code> (Fresh Food / Medicines)	Perishable or urgent orders are more likely to both get delayed AND trigger refunds when quality degrades

(d) Bimodal Pattern in Tier-1 Cities

Operational reasons for two peaks (12–14 min and 28–32 min):

1. **Two types of delivery zones** — Tier-1 cities have dense urban cores (quick deliveries) AND congested arterial roads or high-rises where riders face parking/elevator delays. The same hub serves both.
2. **Peak vs off-peak hours** — Orders during rush hours (8–10 AM, 7–9 PM) consistently take longer. The second peak represents peak-hour orders stacking up.
3. **Order complexity** — Simple grocery orders (few items, pre-packed) vs. Fresh Food or multi-item orders that require more prep, creating two natural prep-time clusters.

Why must this be addressed before modeling? A bimodal distribution violates the assumption that your target variable's predictors behave uniformly. If you treat all orders from a Mumbai hub identically, the model gets confused — a 20-minute delivery is "on-time" for one cluster but "late" for the other.

What modeling mistake could occur if ignored? The model may learn a **single decision boundary** around the mean (~18 min), missing the fact that there are really two sub-populations. This leads to high misclassification in the 20–28 minute zone —

orders that are "on-time for Zone 2 but late for Zone 1" get systematically mislabeled. This is called **Simpson's Paradox** in aggregated data.

(e) Business Trade-Off — Precision vs Recall

Kavya's quote breaks down into two costs:

"30+ min delivery → churn increases significantly" → Missing a truly late delivery (False Negative) = customer lost = high cost

"Flagging too many orders as late risk → unnecessary rider reallocations" → False Positives = wasted operational spend = moderate cost

Answer: It depends on the relative cost — but lean toward Recall with a threshold.

If you prioritize...	You minimize...	You risk...
Recall	Missing true late deliveries	Over-reallocation, higher ops cost
Precision	Unnecessary reallocations	Missing actual delays, customer churn

Best answer: Prioritize **Recall** in the model, but implement a **tiered response system** based on confidence score:

- High probability of late (>0.8) → Immediate rider reallocation
- Medium probability ($0.5-0.8$) → Alert supervisor only
- Low probability (<0.5) → No action

This balances both concerns without over-committing operationally.

(f) Feature Engineering — Three Proposals

1. $\text{relative_delay_ratio} = \text{delivery_time_mins} / \text{prep_time_mins}$

Captures how much of the total time was spent *after* prep — a high ratio signals a rider bottleneck, not a kitchen bottleneck. Directly isolates last-mile delay.

2. $\text{order_complexity_score} = \text{items_count} \times (1 + (\text{order_category} == \text{"Fresh Food"}))$

Fresh Food orders require individual packing — weighting them higher captures the non-linear relationship between item count and prep time.

3. `hub_peak_load_flag = 1` if `(order_hour in [8,9,10,19,20,21]) AND (is_weekend == 0)`
else 0

Flags weekday rush hours when hub throughput is saturated. `delivery_time_mins` spikes during these windows even when `rider_distance_km` is low — this feature isolates that effect cleanly.